Unit 2 Guided Notes — Exponent Rules

Name _____ Date _____ Period _____

Same base? Move the exponents.

🎯 By the end of this unit, I can…

- Apply the product, quotient, and power rules of exponents
- Simplify expressions with negative and zero exponents

Standards: NC.8.EE.1

Fill in the blanks as we go. For every example and problem, **write each step on its own line** in the space given, and box your final answer.

1 Exponent Rules

📖 Words to know

Base: the number being multiplied. In 2^5, the base is 2.

Exponent: the small number up high. It tells how many times to multiply the base. In 2^5, the exponent is 5.

Coefficient: the number in front of a variable. In $4x^3$, the coefficient is 4.

Monomial: one term: a number, a variable, or a product of them, like $4x^3$ or $-2xy^2$.

📌 KEY RULE: Exponent rules only apply when the BASES are the SAME!

Product Rule — Count the Factors

$$x^3 \quad \boxed{x}\boxed{x}\boxed{x}$$
$$\times\, x^2 \quad \boxed{x}\boxed{x}$$
$$= x^5$$

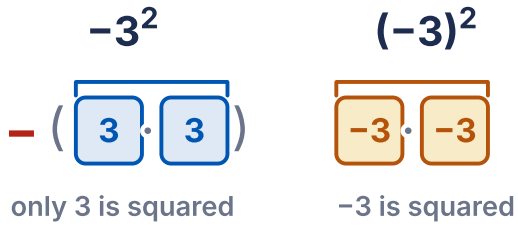
3 + 2 = 5 factors of x

$x^3 \times x^2 = x^{3+2} = x^5$. Count the factors: 3 + 2 = 5 → **ADD** the exponents.

The Five Core Rules

Rule	Formula	Simple Example	Monomial Example	Key Phrase
Product Rule	$x^a \times x^b = x^{a+b}$	_____	_____	_____
Quotient Rule	$x^a \div x^b = x^{a-b}$	_____	_____	_____
Power Rule	$(x^a)^b = x^{a \cdot b}$ (multiply the exponents)	_____	_____	_____
Zero Exponent	$x^0 = 1$	_____	_____	_____
Negative Exponent	$x^{-n} = \frac{1}{x^n}$	_____	_____	_____

⚠️ **PARENTHESES MATTER!** -3^2 = _____ vs. $(-3)^2$ = _____. Without ( ), ONLY the base gets the exponent — not the negative sign!



The little ^{2} belongs to whatever sits right next to it: the 3 alone, or the whole (−3).

Modeling the Rules — See WHY

Each rule comes from writing out the repeated multiplication.

Product Rule · $x^3 \times x^2 = x^5$

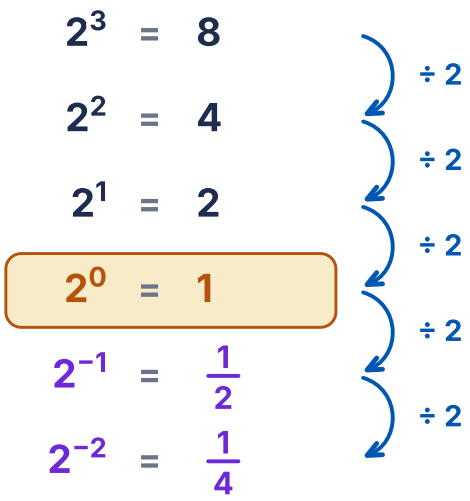
Quotient Rule · $x^5 \div x^2 = x^3$

Power Rule · $(x^2)^3 = x^6$

Zero Exponent · $x^0 = 1$

Negative Exponent · $x^{-3} = \frac{1}{x^3}$

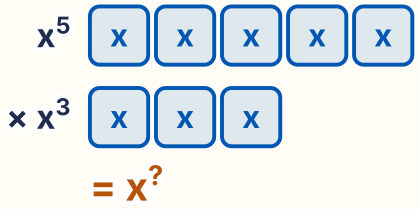
The Pattern Behind x^0 and x^{-n}



Each step down the exponent drops by 1 and the value is **÷ 2**. Keep going past 2^1 and you get $2^0 = 1$, then fractions.

Quick Check

MC Score 3 **Simplify $x^5 \cdot x^3$. Which rule applies, and what is the result?**



Same base, multiplied: count every x tile.

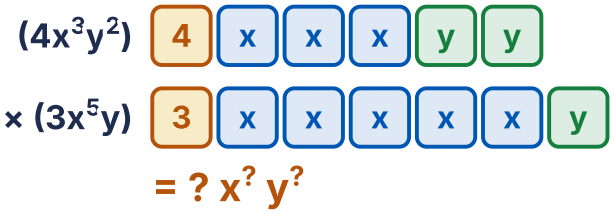
- ○ x^8 (Product Rule)
- ○ x^{15} (Power Rule)
- ○ x^2 (Quotient Rule)

Match the Rule to its Key Phrase

MATCH Score 3 **Write the letter of each rule's key phrase on the line.**

_____ 1. Product Rule	**A.** Power to power? MULTIPLY
_____ 2. Quotient Rule	**B.** Same base? ADD exponents
_____ 3. Power Rule	**C.** Flip to a fraction
_____ 4. Negative Exponent	**D.** Same base? SUBTRACT exponents

Worked Example — Simplify $(4x^3y^2)(3x^5y)$



Multiply the gold numbers. Count the blue x tiles and the green y tiles.

Step 1 — Multiply coefficients: $4 \times 3 = 12$

Step 2 — Add x exponents: $x^3 \cdot x^5 = x^8$

Step 3 — Add y exponents: $y^2 \cdot y^1 = y^3$

✓ **Step 4 — Combine:** $12x^8y^3$

ERROR ANALYSIS Simplify $(3x^2)^3$. A student wrote $3x^6$. What went wrong?

$(3x^2)$ 3 x x

$\times (3x^2)$ 3 x x

$\times (3x^2)$ 3 x x

$(3x^2)^3$ means three copies of $3x^2$. Look at the gold tiles: how many 3s are there?

Part 2 — Practice Problems

📌 **Show every step on its own line.** Box your final answer.

Every problem has a level: Score 3 Score 4 Score 5 . Start with Score 3, then move up.

A. Exponent Rules with Monomials

1) Score 3 **Simplify $x^3 \cdot x^5$**

x^3 [x] [x] [x]

$\times x^5$ [x] [x] [x] [x] [x]

$= x^{?}$

Count all the x tiles.

Answer: _____

2) Score 3 **Simplify $(y^4)^3$**

y^4 [y] [y] [y] [y]

$\times y^4$ [y] [y] [y] [y]

$\times y^4$ [y] [y] [y] [y]

$= y^{?}$

$(y^4)^3$ means 3 copies of y^4.

Answer: _____

3) Score 3 **Simplify $a^8 \div a^3$**

$$\frac{[a][a][a][a][a][a][a][a]}{[a][a][a]}$$

$= a^{?}$

Cancel matching pairs, top and bottom.

Answer: _____

4) Score 4 **Simplify $(2x^3)(5x^2)$**

$(2x^3)$ [2] [x] [x] [x]

$\times (5x^2)$ [5] [x] [x]

$= ?\ x^{?}$

Multiply the numbers; count the x tiles.

Answer: _____

5) Score 4 Simplify $(3m^2)^3$

$(3m^2)$ [3] [m] [m]

$\times (3m^2)$ [3] [m] [m]

$\times (3m^2)$ [3] [m] [m]

$= ?\ m^?$

3 copies: every copy has a 3 and two m tiles.

Answer: _____

6) Score 4 Simplify $(-2a^4)^2$

$(-2a^4)$ [−2] [a] [a] [a] [a]

$\times (-2a^4)$ [−2] [a] [a] [a] [a]

$= ?\ a^?$

2 copies: (−2)(−2) and all the a tiles.

Answer: _____

7) Score 4 Simplify $18n^6 \div 6n^6$

$$\frac{[18]\ [n]\ [n]\ [n]\ [n]\ [n]\ [n]}{[6]\ [n]\ [n]\ [n]\ [n]\ [n]\ [n]}$$

$= ?$

Divide the numbers; cancel matching n pairs.

Answer: _____

8) Score 5 Simplify $(5x^2y)(3xy^3)$

$(5x^2y)$ [5] [x] [x] [y]

$\times (3xy^3)$ [3] [x] [y] [y] [y]

$= ?\ x^?\ y^?$

Multiply the numbers; count x tiles and y tiles.

Answer: _____

9) Score 5 Simplify $(4b^3)^2 \div 2b$

$$\frac{(4)(b)(b)(b)\,(4)(b)(b)(b)}{(2)(b)} = ?\ b^{?}$$

$(4b^3)^2$ is 2 copies of $4b^3$ on top.

Answer: _____

10) Score 5 Simplify $(-3x^2)(-4x^3)$

$(-3x^2)$ $(-3)(x)(x)$

$\times (-4x^3)$ $(-4)(x)(x)(x)$

$= ?\ x^{?}$

Watch the signs: negative × negative.

Answer: _____

11) Score 5 Simplify $(x^5y^2)^3$

(x^5y^2) $(x)(x)(x)(x)(x)(y)(y)$

$\times (x^5y^2)$ $(x)(x)(x)(x)(x)(y)(y)$

$\times (x^5y^2)$ $(x)(x)(x)(x)(x)(y)(y)$

$= x^{?}\ y^{?}$

3 copies of x^5y^2.

Answer: _____